PhonePe Pulse Data Visualization and Exploration

Domain: FinTech | Tool: Power BI | Dataset: PhonePe Pulse

Executive Summary

This project converts PhonePe Pulse transaction and user data into an interactive Power BI dashboard suite. The dashboards support business decision-making by showing transaction trends, user adoption, payment category behavior, state-wise performance, district-wise usage, and geo-spatial digital payment intensity across India.

1. Business Objective

- Analyze PhonePe transaction and user data to identify transaction growth, user growth, regional performance, and payment behavior.
- Build an interactive geo-visualization dashboard with filters for state, district, year, quarter, payment category, amount range, user range, and Top N selections.
- Track business performance using KPIs such as Total Transaction Amount, Total Transaction Count, Total Users, Average Transaction Value, YoY Growth %, and QoQ Growth %.
- Help business users compare high-performing and low-performing regions and understand digital payment adoption patterns across India.

2. Business Problem

- PhonePe Pulse contains a large volume of transaction and user data that is difficult to interpret directly from raw files.
- Therefore, a visual dashboard is required to convert this data into clear, interactive, and decision-ready insights.
- Raw data also may contain Null values, duplicate values and missing data types , it may break ETL process if ETL process will break then it will give wrong data analysis report.
- Business teams need a simple way to identify where transactions are growing, which payment categories are dominant, and which regions have high or low user adoption.
- Without dashboards, decision-makers cannot quickly compare state-wise, district-wise, yearly, and quarterly performance.

3. Why Power BI Was Chosen

- Power BI provides interactive dashboards with slicers, dropdowns, KPI cards, maps, and business-ready visuals.
- It supports quick data modeling and DAX measures such as Total Transaction Amount, Total Transaction Count,YoY Growth %, QoQ Growth %, and Average Transaction Value.
- It is suitable for business users because dashboards are easy to filter by year, quarter, state, district, payment category, and user range.
- Power BI maps make regional analysis easier by showing transaction and user patterns geographically.
- It allows decision-makers to move from static reports to interactive self-service analytics.

4. Data Sources

File Name	Purpose
aggregated_transaction.csv	Overall transaction analysis, payment category distribution, transaction count, and transaction amount.
aggregated_user.csv	Overall user analysis and mobile brand-wise user distribution.
map_transaction.csv	State/district-level transaction count and transaction amount for geo transaction mapping.
map_user.csv	State/district-level registered users for user geo mapping.
top_transaction.csv	Top districts/states by transaction amount and transaction count.
top_user.csv	Top districts/states by registered users.

5. Data Preparation and Modeling

1. Imported all CSV files into Power BI.
2. Checked data types for state, district, year, quarter, transaction count, amount, and registered users.
3. Removed unnecessary index/unnamed columns.
4. Merged map_user and map_transaction into a District_summary table using State, Year, Quarter, and District.
5. Created DAX measures for KPI cards and growth analysis.
6. Built slicers/dropdowns for State, District, Year, Quarter, Transaction Type, Amount Range, User Range, Top N, Growth Filter, and Payment Category.

6. Key KPI Measures

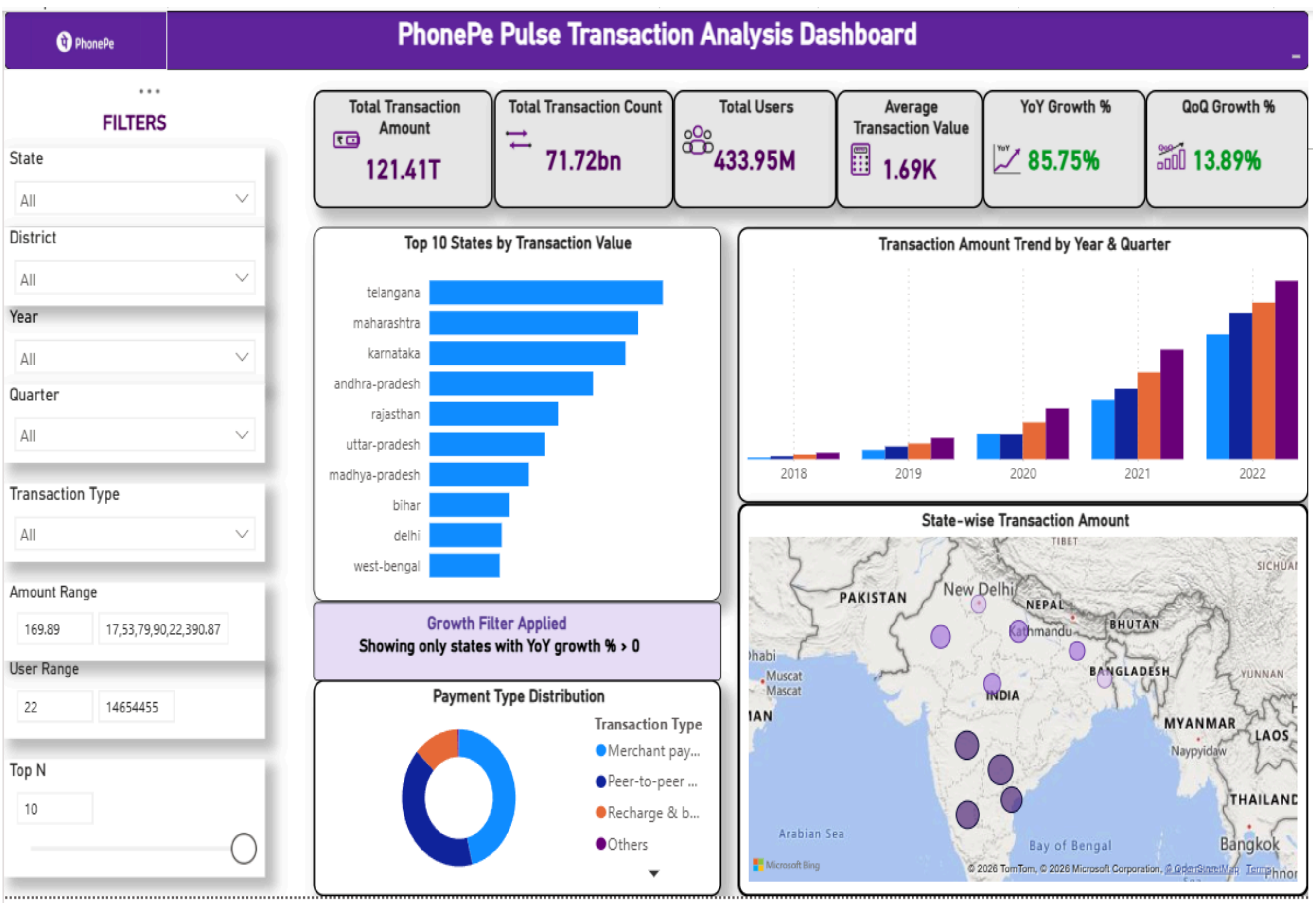
KPI	Business Meaning
Total Transaction Amount	Total value of all PhonePe transactions.
Total Transaction Count	Total number of transactions recorded.
Total Users	Latest/aggregated registered users shown in the dashboard.
Average Transaction Value	Total Transaction Amount divided by Total Transaction Count.
YoY Growth %	Year-on-Year growth in transaction amount.
QoQ Growth %	Quarter-on-Quarter growth in transaction amount.
Highest Revenue District	District with the highest transaction amount.

7. Dashboard Pages Created

7.1 Dashboard 1 - Transaction Analysis

Purpose: To analyze transaction value, transaction count, payment categories, state-wise ranking, growth, and transaction distribution across India.

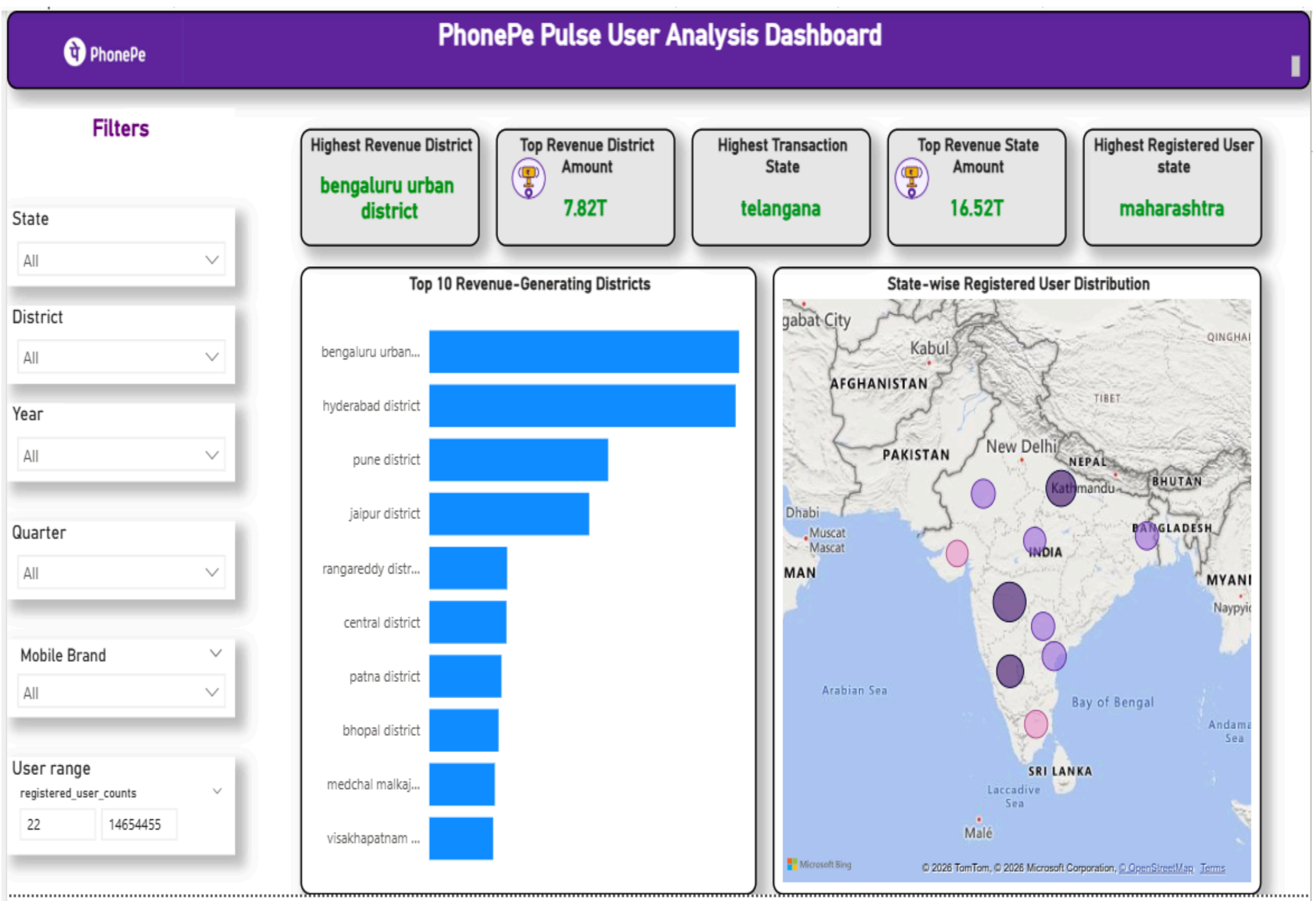
1. KPI cards for transaction amount, transaction count, users, average transaction value, YoY growth, and QoQ growth.
2. Top 10 States by Transaction Value chart.
3. Transaction Amount Trend by Year and Quarter.
4. Payment Type Distribution donut chart.
5. State-wise transaction map and growth filter.



7.2 Dashboard 2 - User Analysis

Purpose: To identify high user adoption states, districts and compare user distribution with transaction value.

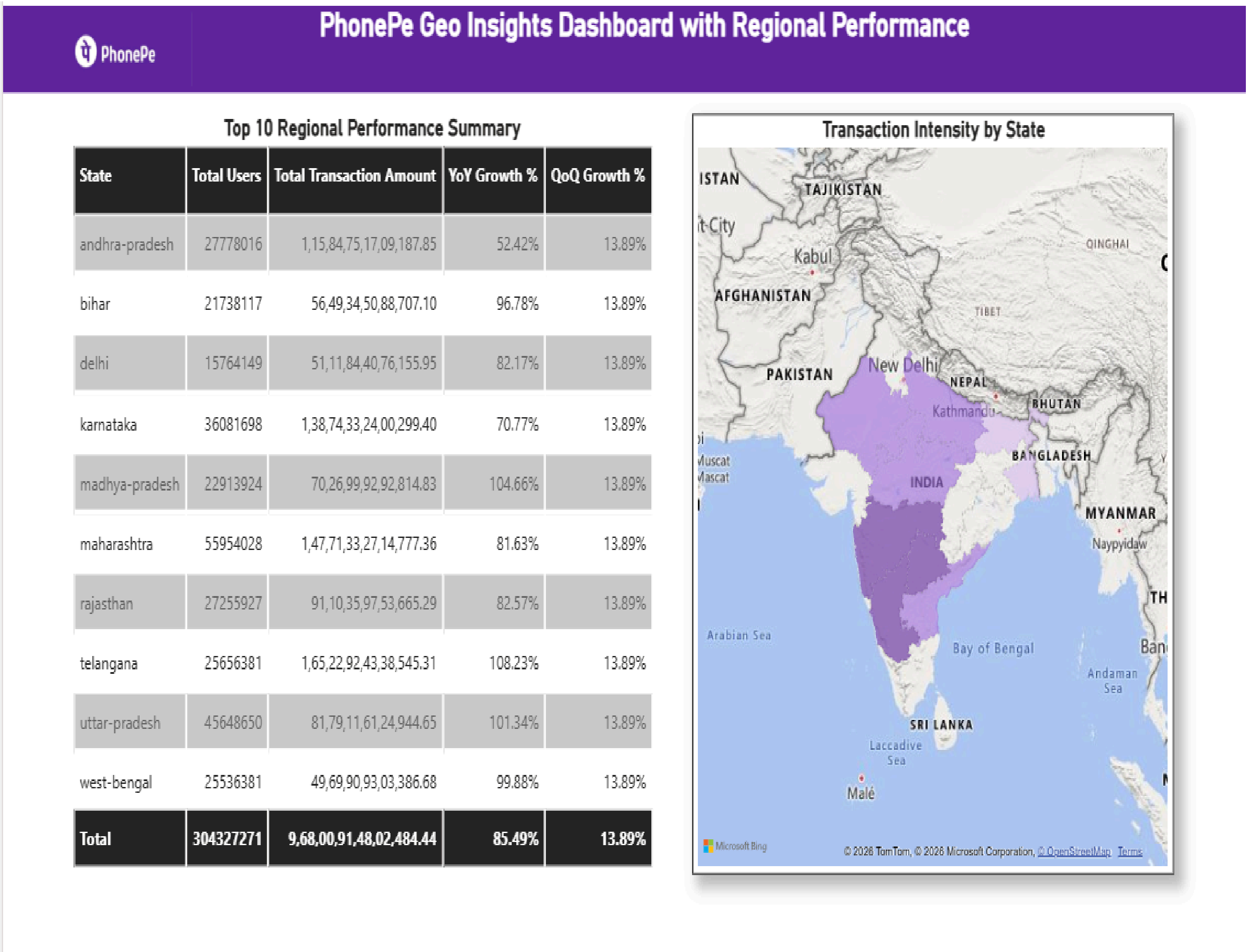
1. Total Users KPI.
2. Highest Revenue District and Top Revenue District Amount KPI cards.
3. Top 10 Revenue-Generating Districts chart.
4. State-wise Registered User Distribution map.
5. This dashboard helps businesses to understand high user regions and also generate high transaction value.



7.3 Dashboard 3 - Geo Performance

Purpose: To visualize regional digital payment performance using geo-based maps and a state-wise performance table.

1. Top 10 Regional Performance Summary table.
2. Transaction Intensity by State filled map.
3. State-wise comparison of total users, transaction amount, and YoY growth.
4. Purple gradient map theme: light purple for low values and dark purple for high values.



State	Total Users	Total Transaction Amount	YoY Growth %	QoQ Growth %
andhra-pradesh	27778016	1,15,84,75,17,09,187.85	52.42%	13.89%
bihar	21738117	56,49,34,50,88,707.10	96.78%	13.89%
delhi	15764149	51,11,84,40,76,155.95	82.17%	13.89%
karnataka	36081698	1,38,74,33,24,00,299.40	70.77%	13.89%
madhya-pradesh	22913924	70,26,99,92,92,814.83	104.66%	13.89%
maharashtra	55954028	1,47,71,33,27,14,777.36	81.63%	13.89%
rajasthan	27255927	91,10,35,97,53,665.29	82.57%	13.89%
telangana	25656381	1,65,22,92,43,38,545.31	108.23%	13.89%
uttar-pradesh	45648650	81,79,11,61,24,944.65	101.34%	13.89%
west-bengal	25536381	49,69,90,93,03,386.68	99.88%	13.89%
Total	**304327271**	**9,68,00,91,48,02,484.44**	**85.49%**	**13.89%**

8. Useful Business Insights

Insight	Business Meaning
Strong transaction growth **(PhonePe transaction value increased strongly over the years.)**	The transaction trend chart shows continuous growth across years and quarters, indicating increasing adoption of digital payments.
High-value states	States such as Telangana, Maharashtra, Karnataka, Andhra Pradesh, Rajasthan, Uttar Pradesh, Madhya Pradesh, Bihar, Delhi, and West Bengal appear in the top transaction ranking. These states contribute strongly to transaction value.
Payment category behavior **(merchant payments are the top transaction type)**	The payment type distribution visual helps identify which categories, such as merchant payments, peer-to-peer payments, recharge and bill payments, financial services, and others, contribute most to platform usage.
Top revenue district	Bengaluru Urban District appears as the highest revenue district in the user analysis dashboard, with a transaction amount around 7.82T.
User concentration **(some districts have high registered users and also strong transaction value, which means good user engagement.)**	The user distribution map shows stronger user concentration in major urban and economically active regions, which indicates stronger digital payment adoption in those areas.
Regional opportunity	Some regions may have high users but relatively lower transaction value. These areas can be targeted for campaigns, merchant onboarding, offers, and user engagement programs.
Geo performance view	The filled map clearly shows transaction intensity by state, helping businesses identify high-performing and low-performing regions quickly.

9. Business Recommendations

1. Based on this analysis, businesses can focus more on high-growth states, promote offers in low-performing regions, and improve merchant payment adoption.

2. Power BI helps businesses by giving interactive dashboards, quick filtering, KPI tracking, and map-based regional analysis.

3. Use top revenue district insights to understand successful adoption patterns and replicate them in lower-performing regions.

4. Target districts with high registered users but lower transaction value using cashback, merchant offers.
5. Use geo maps to plan region-wise expansion, merchant acquisition, and customer engagement strategies.

10. Conclusion

This PhonePe Pulse Data Visualization project successfully converts transaction and user data into an interactive Power BI analytics solution. The dashboards help understand transaction trends, user adoption, payment category behavior, high-revenue districts, and regional digital payment performance. By using Power BI visualizations, businesses can move from raw data to actionable insights and make better decisions for growth, customer engagement, merchant expansion, and regional strategy.